

Final Project Proposal:

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Elevator Pitch:

Develop a CNN model that will classify the difference between different modes of transportation (i.e. cars, trucks, buses, bicycles, trains) in order to train a vision model to detect objects on the road.

Context:

We believe that autonomous vehicles are the future of transportation, or at least minimizing casualties on the road through assisted driving. And given the politicization, bias, and efforts that it takes in order to progress in the motor vehicle industry. That being said, it is important that the public has access to understand, contribute, and critique the models that are being used in an effort to be more transparent with the technologically-enabled vehicles on the road.

Methods:

We wish to use Convolutional Neural Networks in order to classify the objects within the images. Thankfully, we have found a dataset that has labeled pictures from cities, and so we hope to create a classifier using a convolutional neural network. We think that with enough convolutional layers, we can start detecting meaningful enough features to decipher between the different vehicles.

Data Source:

Cityscapes Dataset, which has been split up already as well!

<https://www.cityscapes-dataset.com/dataset-overview/>

Code Resources:

We don't anticipate using any, except for maybe stack overflow in order to debug and see where we can make improvements on our code.

What's new:

Unlike other companies, we are making our project open for the public to use. We will look at many different preprocessing techniques to improve the overall classification accuracy. Finally, we will implement data augmentation to allow our model to be applicable in real-world scenarios.

Plan:

Milestone 1 (March 31)

- Determine which data preprocessing and data augmentation techniques we are going to use.
- Create and design a CNN architecture and train it with the Cityscapes dataset.

Milestone 2 (April 15)

- Improving the CNN model by adjusting hyperparameters
- Evaluate model performance on validation and test datasets, using metrics such as accuracy and F1-score.

Final Submission (April 30)

- Final evaluation of models and testing
- Prepare the final paper and create a presentation.

Proposed

We will measure our model with metrics such as F1 score and accuracy. Understand how the model works and what specific features it looks at when looking at an image. Finally, we can also compare our model results to a model that uses a scikit-learn framework.

Experiments:

We can use data augmentation techniques. We can work with the images and alter specific features such as rotation or brightness of the overall image. We can see how the model reacts to the changes and determine the impact it has.